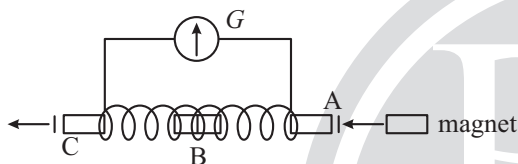


# Electromagnetic Induction

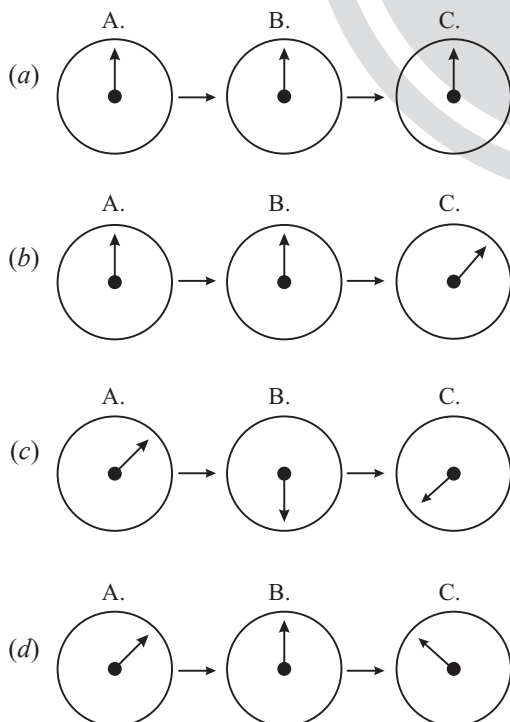
## Single Correct Type Questions

1. A small bar magnet is moved through a coil at constant speed from one end to the other. Which of the following series of observations will be seen on the galvanometer  $G$  attached across the coil? [4 Sep, 2020 (Shift-I)]



Three positions shown describe:

- A. the magnet's entry  
B. magnet is completely inside and  
C. magnet's exit.



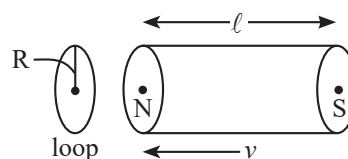
2. A conducting loop of radius  $\frac{10}{\sqrt{\pi}} \text{ cm}$  is placed perpendicular to a uniform magnetic field of  $0.5 \text{ T}$ . The magnetic field is decreased to zero in  $0.5 \text{ s}$  at a steady rate. The induced emf in the circular loop at  $0.25 \text{ s}$  is: [24 Jan, 2023 (Shift-I)]

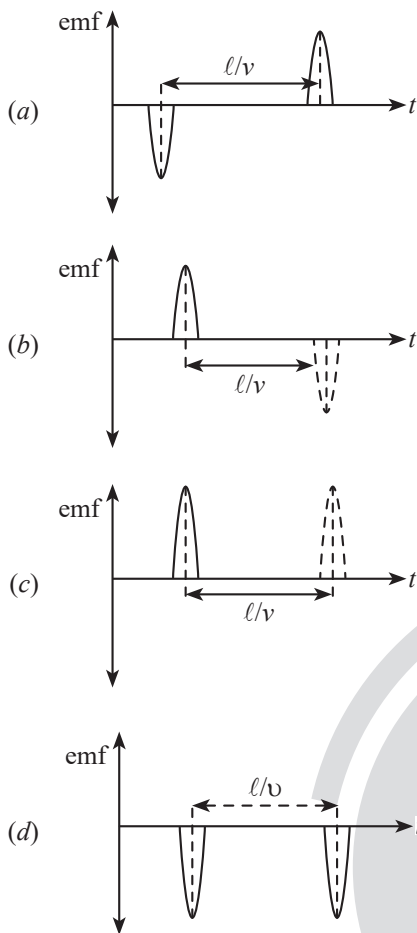
- (a)  $\text{emf} = 1 \text{ mV}$   
(b)  $\text{emf} = 10 \text{ mV}$   
(c)  $\text{emf} = 100 \text{ mV}$   
(d)  $\text{emf} = 5 \text{ mV}$

3. A square loop of area  $25 \text{ cm}^2$  has a resistance of  $10\Omega$ . The square loop is placed in a uniform magnetic field of magnitude  $40.0 \text{ T}$ . The plane of the loop is perpendicular to the magnetic field. The work done in pulling the loop out of the magnetic field slowly and uniformly in  $1.0 \text{ sec}$ , will be [29 Jan, 2023 (Shift-II)]

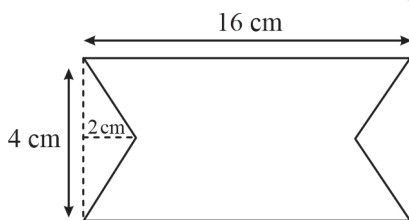
- (a)  $2.5 \times 10^{-3} \text{ J}$   
(b)  $1.0 \times 10^{-3} \text{ J}$   
(c)  $1.0 \times 10^{-4} \text{ J}$   
(d)  $5 \times 10^{-3} \text{ J}$

4. A bar magnet is passing through a conducting loop of radius  $R$  with velocity  $v$ . The radius of the bar magnet is such that it just passes through the loop. The induced e.m.f. in the loop can be represented by the approximate curve: [27 Aug, 2021 (Shift-I)]

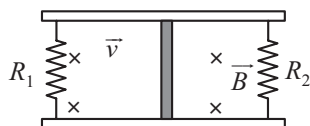




5. At time  $t = 0$  magnetic field of 1000 gauss is passing perpendicularly through the area defined by the closed loop shown in the figure. If the magnetic field reduces linearly to 500 gauss, in the next 5 s, then induced emf in the loop is: [8 Jan, 2020 (Shift-I)]



- (a)  $28 \mu V$  (b)  $36 \mu V$  (c)  $48 \mu V$  (d)  $56 \mu V$
6. A conducting bar of length  $L$  is free to slide on two parallel conducting rails as shown in the figure. [16 March, 2021 (Shift-I)]



Two resistors  $R_1$  and  $R_2$  are connected across the ends of the rails. There is a uniform magnetic field  $\vec{B}$  pointing into the page. An external agent pulls the bar to the left at a constant speed  $v$ .

The correct statement about the directions of induced currents  $I_1$  and  $I_2$  flowing through  $R_1$  and  $R_2$  respectively is

- (a)  $I_1$  is in anticlockwise direction and  $I_2$  is in clockwise direction  
 (b) Both  $I_1$  and  $I_2$  are in anticlockwise direction  
 (c) Both  $I_1$  and  $I_2$  are in clockwise direction  
 (d)  $I_1$  is in clockwise direction and  $I_2$  is in anticlockwise direction

7. A wire of length 1m is moving with a velocity of 8m/s at right angles to a magnetic field of 2T. The magnitude of induced emf, between the ends of wire the will be

[25 Jan, 2023 (Shift-II)]

- (a) 20 V (b) 8 V (c) 12 V (d) 16 V

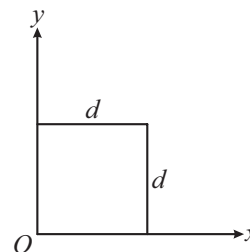
8. An emf of 0.08 V is induced in a metal rod of length 10 cm held normal to a uniform magnetic field of 0.4 T, when moves with a velocity of: [8 April, 2023 (Shift-II)]

- (a)  $2 \text{ ms}^{-1}$   
 (b)  $3.2 \text{ ms}^{-1}$   
 (c)  $0.5 \text{ ms}^{-1}$   
 (d)  $20 \text{ ms}^{-1}$

9. The magnetic field in a region is given by  $\vec{B} = B_0 \left( \frac{x}{a} \right) \hat{k}$ .

A square loop of side  $d$  is placed with its edges along the  $x$  and  $y$  axes. The loop is moved with a constant velocity  $\vec{v} = v \hat{i}$ . The emf induced in the loop is:

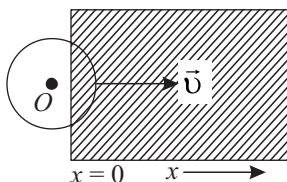
[16 March, 2021 (Shift-II)]



- (a)  $\frac{B_0 v_0 d^2}{a}$  (b)  $\frac{B_0 v_0}{2a}$   
 (c)  $\frac{B_0 v_0^2 d}{2a}$  (d)  $\frac{B_0 v_0 d^2}{2a}$

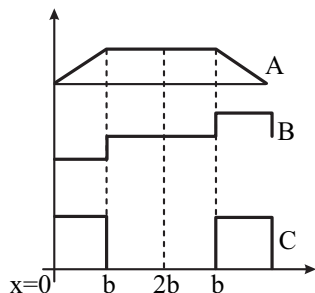
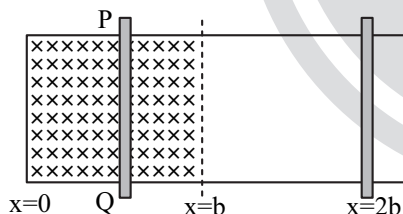
10. A constant magnetic field of  $1\text{ T}$  is applied in the  $x > 0$  region. A metallic circular ring of radius  $1\text{ m}$  is moving with a constant velocity of  $1\text{ m/s}$  along the  $x$ -axis. At  $t = 0$ , the centre of  $O$  of the ring is at  $x = -1\text{ m}$ . What will be the value of the induced emf in the ring at  $t = 1\text{ s}$ ? (Assume the velocity of the ring does not change.)

[27 Aug, 2021 (Shift-II)]



- (a)  $0\text{ V}$   
 (b)  $2\pi\text{ V}$   
 (c)  $2\text{ V}$   
 (d)  $1\text{ V}$
11. The arm  $PQ$  of a rectangular conductor is moving from  $x = 0$  to  $x = 2b$  outwards and then inwards from  $x = 2b$  to  $x = 0$  as shown in the figure. A uniform magnetic field perpendicular to the plane is acting from  $x = 0$  to  $x = b$ . Identify the graph the variation of different quantities with distance.

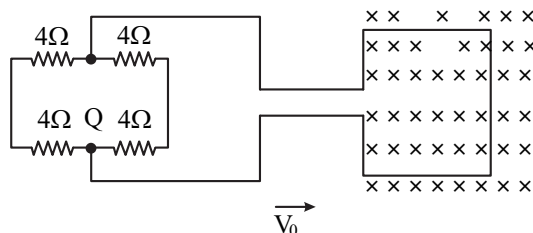
[20 July, 2021 (Shift-I)]



- (a) A – Flux, B – emf, C – Power dissipated  
 (b) A – emf, B – Power dissipated, C – Flux  
 (c) A – Power dissipated, B – Flux, C – emf  
 (d) A – Flux, B – Power dissipated, C – emf

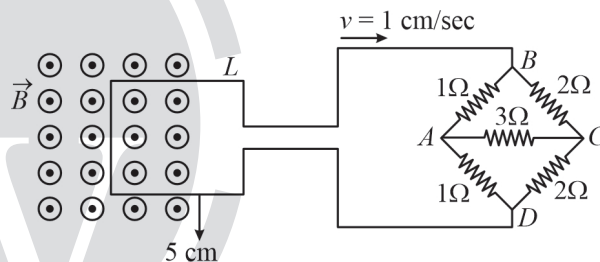
12. A square loop of side  $20\text{ cm}$  and resistance  $1\Omega$  is moved towards right with a constant speed  $v_0$ . The right arm of the loop is in a uniform magnetic field of  $5\text{ T}$ . The field is perpendicular to the plane of the loop and is going into it. The loop is connected to a network of resistors each of value  $4$ . What should be the value of  $v_0$  so that a steady current of  $2\text{ mA}$  flows in the loop

[1 Sep, 2021 (Shift-II)]



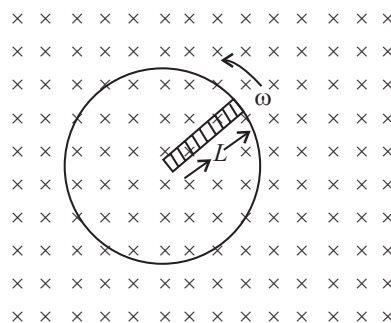
- (a)  $1\text{ m/s}$   
 (b)  $10^2\text{ m/s}$   
 (c)  $10^{-2}\text{ cm/s}$   
 (d)  $1\text{ cm/s}$
13. The figure shows a square loop  $L$  of side  $5\text{ cm}$  which is connected to a network of resistances. The whole setup is moving towards right with a constant speed of  $1\text{ cm/s}$ . At some instant, a part of  $L$  is in a uniform magnetic field of  $1\text{ T}$ , perpendicular to the plane of the loop. If the resistance of  $L$  is  $1.7\Omega$ , the current in the loop at that instant will be close to

[12 April, 2019 (Shift-I)]



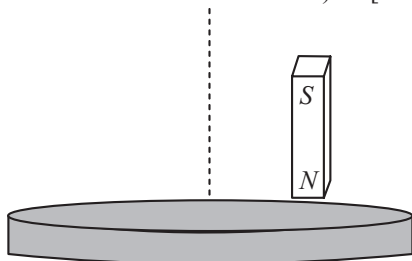
- (a)  $115\text{ }\mu\text{A}$   
 (b)  $170\text{ }\mu\text{A}$   
 (c)  $60\text{ }\mu\text{A}$   
 (d)  $150\text{ }\mu\text{A}$
14. A metallic rod of length ' $L$ ' is rotated with an angular speed of ' $\omega$ ' normal to a uniform magnetic field ' $B$ ' about an axis passing through one end of rod as shown in figure. The induced emf will be:

[24 Jan, 2023 (Shift-II)]

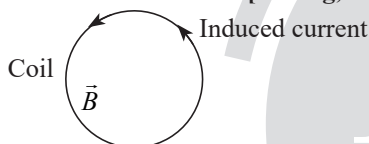


- (a)  $\frac{1}{4}B^2L\omega$   
 (b)  $\frac{1}{4}BL^2\omega$   
 (c)  $\frac{1}{2}BL^2\omega$   
 (d)  $\frac{1}{2}B^2L^2\omega$

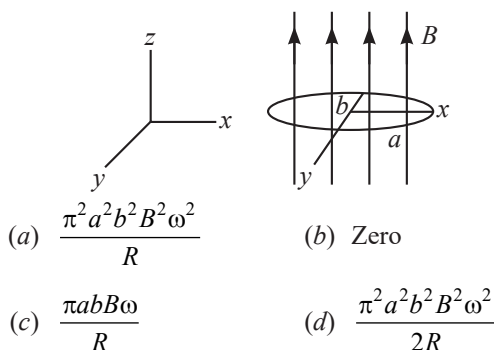
15. A light disc made of aluminium (a nonmagnetic material) is kept horizontally and is free to rotate about its axis as shown in the figure. A strong magnet is held vertically at a point above the disc away from its axis. On revolving the magnet about the axis of the disc, the disc will (figure is schematic and not drawn to scale) [JEE Adv, 2020]



- (a) Rotate in the direction opposite to the direction of magnet's motion.  
 (b) Rotate in the same direction as the direction of magnet's motion.  
 (c) Not rotate and its temperature will remain unchanged.  
 (d) Not rotate but its temperature will slowly rise.
16. A coil is placed in a magnetic field as shown below:  $\vec{B}$  is:

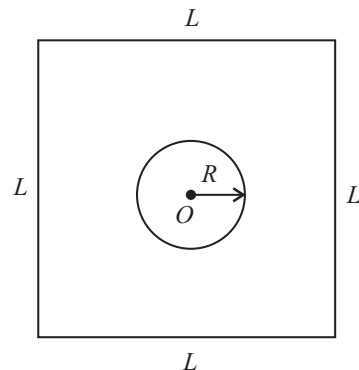


- (a) Outward and decreasing with time  
 (b) Parallel to the plane of coil and decreasing with time  
 (c) Parallel to the plane of coil and increasing with time  
 (d) Outward and increasing with time
17. An elliptical loop having resistance  $R$ , of semi major axis  $a$  and semi minor axis  $b$  is placed in a magnetic field as shown in the figure. If the loop is rotated about the  $x$ -axis with angular frequency  $\omega$ , the average power loss in the loop due to joule heating is: [3 Sep, 2020 (Shift-I)]



18. Find the mutual inductance of the arrangement, when a small circular loop of wire of radius ' $R$ ' is placed inside a large square loop of wire of side  $L$  ( $L \gg R$ ). The loops are coplanar and their centres coincide:

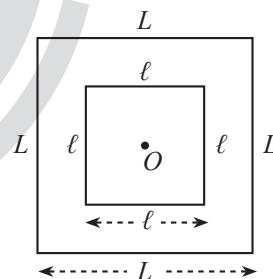
[29 Jan, 2023 (Shift-I)]



- (a)  $M = \frac{\sqrt{2}\mu_0 R^2}{L}$   
 (b)  $M = \frac{2\sqrt{2}\mu_0 R}{L^2}$   
 (c)  $M = \frac{2\sqrt{2}\mu_0 R^2}{L}$   
 (d)  $M = \frac{\sqrt{2}\mu_0 R}{L^2}$

19. A small square loop of wire of side  $l$  is placed inside a large square loop of wire.  $L$  ( $L \gg l$ ). Both loops are coplanar and their centres coincide at point  $O$  as shown in figure. The mutual inductance of the system is:

[25 July, 2022 (Shift-I)]



- (a)  $\frac{2\sqrt{2}\mu_0 L^2}{\pi l}$   
 (b)  $\frac{\mu_0 l^2}{2\sqrt{2}\pi L}$   
 (c)  $\frac{2\sqrt{2}\mu_0 l^2}{\pi L}$   
 (d)  $\frac{\mu_0 L^2}{2\sqrt{2}\pi l}$
20. The time taken for the magnetic energy to reach 25% of its maximum value, when a solenoid of resistance  $R$ , and inductance  $L$  is connected to a battery, is:

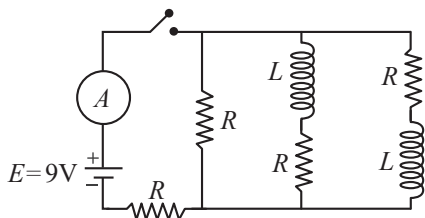
[18 March, 2021 (Shift-II)]

- (a)  $\frac{L}{R} \ln(2)$   
 (b)  $\frac{L}{R} \ln(10)$   
 (c) Infinite  
 (d)  $\frac{L}{R} \ln(5)$

21. Figure shows a circuit that contains four identical resistors with resistance  $R = 2.0\Omega$ , two identical inductors with inductance  $L = 2.0\text{ mH}$  and an ideal battery with emf  $E = 9\text{ V}$ . The current ' $i$ ' just after switch ' $S$ ' is closed will be:

[24 Feb, 2021 (Shift-II)]

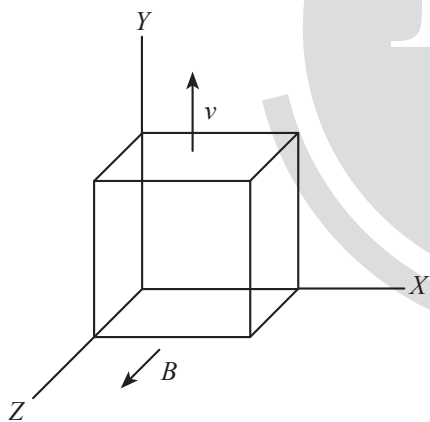
- (a) 2.25 A (b) 3.0 A (c) 9 A (d) 3.37 A



### Integer Type Questions

22. A metallic cube of side 15 cm moving along y-axis at a uniform velocity of  $2\text{ ms}^{-1}$ . In a region of uniform magnetic field of magnitude  $0.5\text{ T}$  directed along z-axis. In equilibrium the potential difference between the faces of higher and lower potential developed because of the motion through the field will be \_\_\_\_\_ mV.

[11 April, 2023 (Shift-II)]



23. A conducting circular loop is placed in  $X - Y$  plane in presence of magnetic field  $\vec{B} = (3t^3\hat{j} + 3t^2\hat{k})$  in SI unit. If the radius of the loop is 1 m, the induced emf in the loop, at time,  $t = 2\text{ s}$  is  $n\pi\text{ V}$ . The value of  $n$  is \_\_\_\_\_.

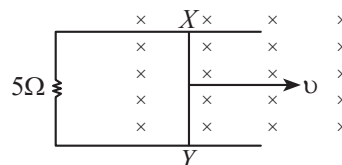
[27 July, 2022 (Shift-II)]

24. In a coil of resistance  $8\Omega$ , the magnetic flux due to an external magnetic field varies with time as  $\phi = \frac{2}{3}(9 - t^2)$ . The value of total heat produced in the coil, till the flux becomes zero, will be \_\_\_\_\_ J.

[26 July, 2022 (Shift-II)]

25. A 1 m long metal rod  $XY$  completes the circuit as shown in figure. The plane of the circuit is perpendicular to the magnetic field of flux density  $0.15\text{ T}$ . If the resistance of the circuit is  $5\Omega$ , the force needed to move the rod in direction, as indicated, with a constant speed of  $4\text{ m/s}$  will be \_\_\_\_\_  $\times 10^{-3}\text{ N}$

[10 April, 2023 (Shift-I)]



26. In an ac generator, a rectangular coil of 100 turns each having area  $14 \times 10^{-2}\text{ m}^2$  is rotated at 360 rev/min about an axis perpendicular to a uniform magnetic field of magnitude  $3.0\text{ T}$ . The maximum value of the emf produced will be \_\_\_\_\_ V. (Take  $\pi = \frac{22}{7}$ )

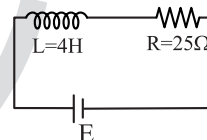
[30 Jan, 2023 (Shift-II)]

27. A conducting circular loop is placed in a uniform magnetic field of  $0.4\text{ T}$  with its plane perpendicular to the field. Somehow, the radius of the loop starts expanding at a constant rate of  $1\text{ mm/s}$ . The magnitude of induced emf in the loop at an instant when the radius of the loop is  $2\text{ cm}$  will be \_\_\_\_\_  $\mu\text{V}$ .

[12 April, 2023 (Shift-I)]

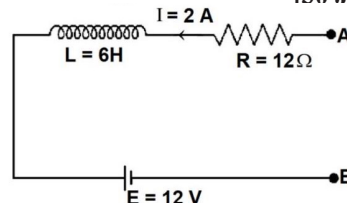
28. In the given figure, an inductor and a resistor are connected in series with a battery of emf  $E$  volt.  $\frac{E^a}{2b}\text{ J/s}$  represents the maximum rate at which the energy is stored in the magnetic field (inductor). The numerical value of  $\frac{b}{a}$  will be \_\_\_\_\_.

[13 April, 2023 (Shift-I)]



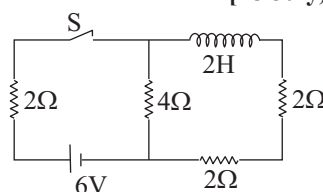
29. As per the given figure, if  $\frac{dI}{dt} = -1\text{ A/s}$  then the value of  $V_{AB}$  at this instant will be \_\_\_\_\_ V.

[30 Jan, 2023 (Shift-I)]



30. For the given circuit the current through battery of  $6\text{ V}$  just after closing the switch ' $S$ ' will be \_\_\_\_\_ A.

[28 July, 2022 (Shift-II)]



## ANSWER KEY

- |         |           |          |         |          |            |          |          |          |         |
|---------|-----------|----------|---------|----------|------------|----------|----------|----------|---------|
| 1. (d)  | 2. (b)    | 3. (b)   | 4. (a)  | 5. (d)   | 6. (d)     | 7. (d)   | 8. (a)   | 9. (a)   | 10. (c) |
| 11. (a) | 12. (d)   | 13. (b)  | 14. (c) | 15. (b)  | 16. (a)    | 17. (d)  | 18. (c)  | 19. (c)  | 20. (a) |
| 21. (a) | 22. [150] | 23. [12] | 24. [2] | 25. [18] | 26. [1584] | 27. [50] | 28. [25] | 29. [30] | 30. [1] |

## EXPLANATIONS

1. (d) Current direction reverses when magnet exits from solenoid and when magnet is completely inside, current is zero.

2. (b) Using Faraday's Law of EMI,  $EMF = \frac{d\phi}{dt} = \frac{BA-0}{t}$

$$A = \pi r^2 = \pi \left( \frac{0.1^2}{\pi} \right) = 0.01 \text{ m}^2 \text{ and } B = 0.5 \text{ T}$$

$$\Rightarrow EMF = \frac{(0.5)(0.01)}{0.5} = 10 \text{ mV}$$

3. (b) Length of each side of the loop,  $\ell = 5 \text{ cm}$   
and,  $t = 1 \text{ sec}$

$$\therefore v = \frac{0.05}{1} = 0.05 \text{ m/s}$$

$$i = \frac{B\ell v}{R} = \frac{40 \times 0.05 \times 0.05}{10} = 0.01 \text{ A}$$

$$\therefore F = i\ell B = 40 \times 0.01 \times 0.05 = 0.02 \text{ N}$$

$$\therefore W = 0.02 \times \ell = 0.02 \times 0.05 = 1 \times 10^{-3} \text{ J}$$

4. (a) According to the faraday's law of electromagnetic induction,  $EMF = \frac{-d\phi}{dt}$

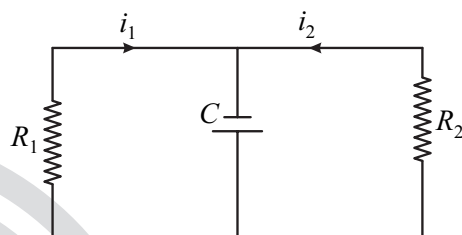
When magnet passes through the centre region of loop, no emf is induced. By using the Lenz's law, the polarity of emf will be negative while entering the loop.

5. (d)  $\varepsilon = \left| -\frac{d\phi}{dt} \right| = \left| -\frac{A dB}{dt} \right|$

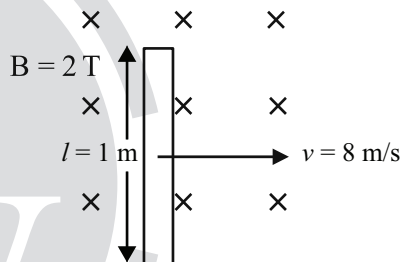
$$= (16 \times 4 - 4 \times 2) \frac{(1000 - 500)}{5} \times 10^{-4} \times 10^{-4}$$

$$= 56 \times \frac{500}{5} \times 10^{-8} = 56 \times 10^{-6} \text{ V}$$

6. (d)



7. (d)



Induced emf across the ends of the rod  $= B\ell v$   
 $= 2 \times 8 \times 1 = 16 \text{ V}$

8. (a) Given length of rod,  $l = 10 \text{ cm} = 0.10 \text{ m}$

Magnetic field,  $B = 0.4 \text{ T}$

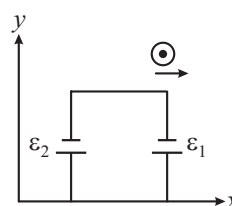
e.m.f.  $\varepsilon = 0.08 \text{ V}$

Induced emf  $\varepsilon = Blv$

$$\therefore v = \frac{\varepsilon}{Bl}$$

$$\Rightarrow v = \left( \frac{0.08}{0.4 \times 0.10} \right) = 2 \text{ m/s}$$

9. (a)



[As  $\vec{v} \times \vec{B}$  is along the conductor length for two arms of the loop]

Assumed after some time, loop moved  $x$  distance.

$$\varepsilon_1 = \frac{B_0(x+d)}{a} v_0 d$$

$$\varepsilon_2 = \frac{B_0 \times d v_0}{a}$$

$$\therefore \varepsilon_{\text{net}} = \varepsilon_1 - \varepsilon_2 = \frac{B_0 v_0 d^2}{a}$$

10. (c) The induced emf at  $t = 1$  s i.e. at  $x = 0$  m is,

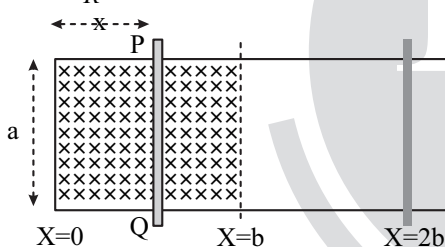
$$E = B l v = B(2R)v$$

$$\Rightarrow E = 1 \times 1 \times 2 \times 1 = 2\text{V}$$

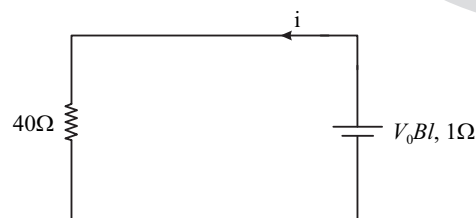
11. (a) When arm  $PQ$  of a rectangular conductor is moving from  $x = 0$  to  $x = b$ , the flux ( $\phi = ax$ ) linked with loop increases and while moving from  $x = b$  to  $x = 2b$  and from  $x = 2b$  to  $x = b$ , flux ( $\phi = ab$ ) remains constant and then flux ( $\phi = ax$ ) decreases to zero as it moves from  $x = b$  to  $x = 0$

$$e_{\text{in}} = -\frac{d\phi}{dt} = -a \frac{dx}{dt} = -av \Rightarrow \text{Induced emf}$$

$$P = \frac{(e_{\text{in}})^2}{R}$$



12. (d)  $I = \frac{V_0 B \ell}{R} = \frac{V_0 \times 5 \times 20 \times 10^{-2}}{4 + 1}$



$$V_0 = \frac{2 \times 10^{-3}}{20 \times 10^{-2}} = 1 \times 10^{-2} \text{ m/s}$$

$$V_0 = 1 \text{ cm/s}$$

13. (b) Since it is a balanced wheatstone bridge, its equivalent resistance =  $\frac{4}{3} \Omega$

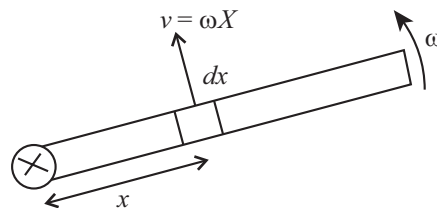
$$\varepsilon = B \ell v = 5 \times 10^{-4} \text{ V}$$

so total resistance

$$R = \frac{4}{3} + 1.7 \approx 3 \Omega$$

$$i = \frac{\varepsilon}{R} \approx 166 \mu\text{A} \approx 170 \mu\text{A}$$

14. (c)



Emf induced in the rod is

$$\int d\varepsilon = \int B(\omega x) dx$$

$$\therefore \varepsilon = B\omega \int_0^L x dx = \frac{B\omega L^2}{2}$$

15. (b) When the magnet is moved, it creates a state where the plate moves through the magnetic flux, due to which an electromotive force is generated in the plate and eddy currents are induced. These currents are such that it opposes the relative motion  $\Rightarrow$  disc will rotate in the direction of rotation of magnet.

**Note:** This apparatus is called Arago's disk.

16. (a) Using Faraday's law  $\vec{B}$  should be outward and decreasing for anticlockwise induced current.

17. (d) Induced emf for rotating loop,

$$\varepsilon_0 = BA\omega = B(\pi ab)\omega \quad [\because A = \pi ab]$$

$\therefore$  Average power loss due to Joule's heating effect is

$$<P> = \frac{\varepsilon_0^2}{2R} = \frac{\pi^2 a^2 b^2 B^2 \omega^2}{2R}$$

18. (c)  $\phi = Mi$  ... (i)

$$\text{and } \phi = (BA) \quad \dots (ii)$$

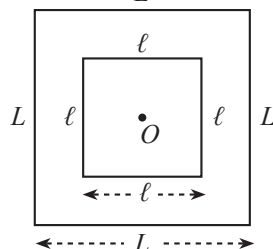
Magnetic field at point O due to the one side of the

$$\text{square loop } B = \frac{\mu_0}{4\pi} \frac{i\sqrt{2}}{\left(\frac{L}{2}\right)}$$

$$\Rightarrow \phi = A \times B_{\text{Total}}$$

$$\Rightarrow \phi = \pi R^2 \left[ 4 \frac{\mu_0}{4\pi} \frac{i}{\left(\frac{L}{2}\right)} \sqrt{2} \right] \Rightarrow M = \frac{2\sqrt{2}\mu_0 R^2}{L}$$

19. (c)





Assuming current  $i$  in outer loop magnetic field at centre, flux through the smaller area

$$\phi = 4 \times \frac{\mu_0}{4\pi} \frac{i}{L/2} \times [\sin 45^\circ + \sin 45^\circ] \times A$$

$$= \frac{2\sqrt{2}\mu_0 i}{\pi L} \ell^2$$

comparing with,  $\phi = Mi$

$$M = \frac{2\sqrt{2}\mu_0 \ell^2}{\pi L}$$

20. (a) Magnetic energy,  $U = \frac{1}{2} LI_0^2$

Given:  $U = 25\%$  of  $U_0$

$$\Rightarrow \frac{1}{2} LI^2 = \frac{1}{4} \times \frac{1}{2} LI_0^2$$

$$\Rightarrow I^2 = \frac{I_0^2}{4} \Rightarrow I = \frac{I_0}{2}$$

$$\therefore I = I_0 (1 - e^{-t/T})$$

$$\Rightarrow \frac{I_0}{2} = I_0 (1 - e^{-t/\tau})$$

$$\Rightarrow \frac{1}{2} = e^{-t/\tau}$$

$$\Rightarrow e^{t/\tau} = 2$$

$$\Rightarrow t = \tau \ln 2$$

$$\Rightarrow t = \frac{L}{R} \ln 2$$

21. (a) We have given,  $R = 2\Omega$ ,  $L = 2 \text{ mH}$ ,  $E = 9 \text{ V}$

Under steady state inductor behaves like a short circuit

$$i = \frac{E}{2R} = \frac{9V}{4\Omega} = 2.25 \text{ A}$$

Just after the switch  $S$  is closed, the inductor acts as open circuit.

22. [150]  $F = qVB$

$$qE = qVB$$

$$\frac{\Delta V}{d} = VB$$

$$\Delta V = (vB)d$$

$$\Delta V = (2 \times 1/2) 0.15$$

$$\Delta V = 150 \text{ mV}$$

23. [12]  $\phi = \vec{B} \cdot \vec{A} = (3t^3 \hat{j} + 3t^2 \hat{k}) \cdot (\pi(1)^2 \hat{k})$

$$\phi = 3t^2 \pi$$

$$\epsilon_{\text{IND}} = \left| \frac{d\phi}{dt} \right| = 6t\pi$$

$$\text{At, } t = 2, \epsilon_{\text{IND}} = 12\pi$$

The value of  $n$  is 12.

24. [2]  $\phi = \frac{2}{3}(9 - t^2) = 0$

$\Rightarrow t = 3 \text{ sec}$  is the when flux becomes zero.

$$emf = \frac{-d\phi}{dt} = \frac{4t}{3}$$

$$I = \frac{emf}{R} = \frac{\frac{4}{3}t}{8} = \frac{t}{6}$$

$$\text{Heat produced} = \int I^2 R dt = \int_0^3 \frac{t^2}{36} \times 8 dt = 2J$$

25. [18] Force needed to move the rod is  $F = i\ell B$

$$= \left( \frac{vB\ell}{R} \right) \ell B = \frac{vB^2 \ell^2}{R} = \frac{4}{5} \times \left( \frac{15}{100} \right)^2 \times 1^2$$

$$F = \frac{4}{5} \times \frac{225}{10^4} = \frac{180}{10^4} = 0.018 \text{ N} = 18 \times 10^{-3} \text{ N}$$

26. [1584]  $Emf$  induced in a rectangular coil is

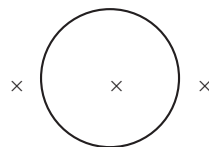
$$\epsilon = NAB\omega \sin \omega t$$

For  $emf$  to be maximum  $\sin \omega t = 1$

$$\Rightarrow \epsilon_{\text{max}} = NBA\omega$$

$$= 100 \times 14 \times 10^{-2} \times 3 \times \frac{360 \times 2\pi}{60} = 1584 \text{ V}$$

27. [50]  $\times$   $\times$



$$\times \quad \times \quad \times$$

$$\frac{dr}{dt} = 10^{-3} \text{ m/s}$$

Area of the loop  $A = \pi r^2$

$$\therefore \frac{dA}{dt} = 2\pi r \frac{dr}{dt}$$

$$\epsilon = \left| \frac{-d\phi}{dt} \right| = \left| \frac{BdA}{dt} \right| = B \cdot 2\pi r \frac{dr}{dt}$$

$$= 0.4 \times 2 \times \pi \times 2 \times 10^{-2} \times 10^{-3} \text{ V}$$

$$= 16 \pi \mu \text{V} = 50.24 \mu \text{V}$$



28. [25]  $E = \frac{1}{2}LI^2 \Rightarrow \frac{dE}{dt} = \frac{1}{2} \times 2LI \frac{dI}{dt}$   
 Rate of energy storing  $= LI \frac{dI}{dt}$

For  $R-L$  circuit

$$I = \frac{E}{R} \left( 1 - e^{-\frac{Rt}{L}} \right)$$

$$\text{So } \frac{dI}{dt} = \frac{E}{L} e^{-\frac{Rt}{L}}$$

$$\frac{dE}{dt} = \frac{E^2}{R} \left( 1 - e^{-\frac{Rt}{L}} \right) \left( e^{-\frac{Rt}{L}} \right)$$

Time at which rate of power storing will be max,

$$t = \frac{L}{R \ln 2}$$

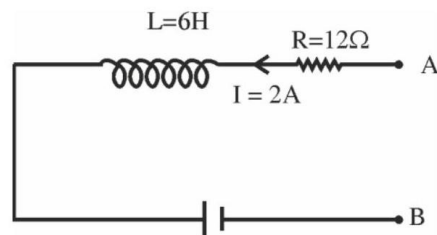
$$\text{So, } \frac{dE}{dt} = \frac{E^2}{R} \left( 1 - \frac{1}{2} \right) \times \frac{1}{2}$$

$$\Rightarrow \frac{E^2}{4R} = \frac{E^2}{100} = \frac{E^2}{2 \times 50}$$

$$a = 2, b = 50$$

$$\text{So, } \frac{b}{a} = 25$$

29. [30]



$$\frac{dI}{dt} = -1 \frac{\text{A}}{\text{sec}}$$

$$V_A - IR - L \frac{dI}{dt} - 12 = V_B$$

$$V_A - 2 \times 12 - 6(-1) - 12 = V_B$$

$$V_A - V_B = 36 - 6 = 30 \text{ volt}$$

30. [1]  $i = \frac{V_{\text{net}}}{R_{\text{net}}} = \frac{6}{6} = 1 \text{ A}$

